The Measurement of Coevolution in Nature

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Introduction

The spatial correlation of phenotypes across a metapopulation has often been cited as evidence for the existence of coevolution between a pair of interacting species[1]. However, recently developed theory has shown that correlations are neither necessary nor sufficient for the existence of a coevolutionary process[2]. This issue motivates the development of a new suite of coevolutionary inference methods that tie theory to data. Here we initiate this development by presenting a relatively simple likelihood-based method that uses easily collected phenotypic data to infer the strength and significance of coevolution.

Potentially Coevolved Systems

Pictured on the left is the camellia-weevil system (*Curculio camellia* and *Camellia japonica*). The weevil bores a hole into the thick pericarp of the camellia fruit to oviposit her eggs. This has been thought to lead to a coevolutionary arms race in weevil rostrum and pericarp thickness[3]. On the right is the fly-flower pollination-based interaction (*Moegistorhynchus Longirostris* and *Lapeirousia anceps*) where it is thought that the proboscis length of the fly has coevolved with the nectar-tube length of the flower[4]. Here we test for coevolution in these two systems.

The Model

$$\mu_i' = \mu_i + G_i\left[B_i\delta_i + B_i(\mu_j - \mu_i) + A_i(\theta_i - \mu_i)\right]$$

$$V_i' = (1 - 2A_iG_i)V_i + 2B_jG_j(C - V_i) + G_i/n_i$$

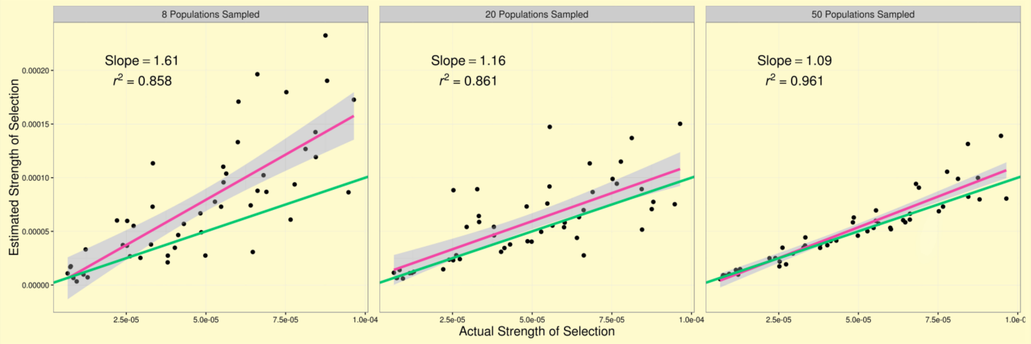
$$C' = (1 - A_1G_1 - A_2G_2)C + B_2G_2(V_1 - C) + B_1G_1(V_2 - C)$$

We develop a model that captures abiotic selection, biotic selection and drift for a pair of interacting species across a metpopulation. In particular, we track the means, variances and spatial covariance of local population trait values. This allows the comparison of expected statistical moments to the observed moments, a necessary component of likelihood-based inference.

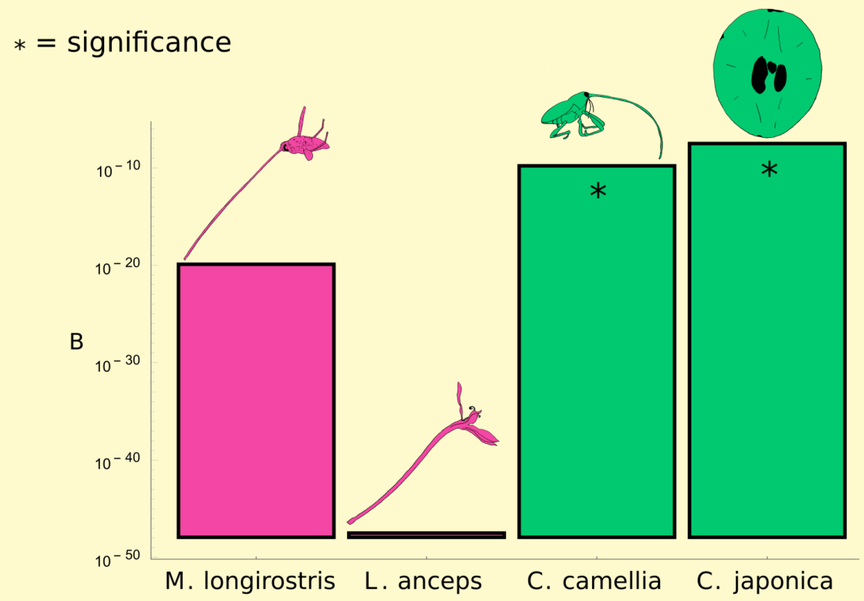
Inference

Assuming equilibrium, we maximize likelihood of the data with respect to model parameters to produce estimates of the strengths of biotic selection. Likelihood ratio statistics are then computed to determine the statistical significance of these estimates.

Method Evaluation



Results



Conclusion

We find strong support for coevolution in the camellia-weevil antagonism but no support for coevolution in the plant-pollinator system. This discrepancy in results may have to do with the generality of the pollinator interaction. With various other species involved with the interaction in an analogous manner, the signal of pairwise coevolution between the fly and the flower has likely been washed away by a diffusive coevolutionary force imposed by the collective interactions of two guilds.

References

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